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Stroke Rehabilitation

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SUMMARY PEARLS

- One in four people who survive a stroke will have a recurrence within 5 years, with the highest recurrence rate within the first 30 days.
- Ischemic strokes account for 85% to 90% of strokes in the United States, with nearly 800,000 occurring annually.
- Overcorrection of blood pressure resulting in arterial hypotension leads to worse outcomes.
- Early mobilization within the first 24 hours of intracerebral hemorrhage may increase mortality; however, it is safe to begin mobilization after 24 to 48 hours.
- Prophylactic treatment for seizures is not indicated, but treatment should be initiated in patients with electroencephalogram confirmed or clinically observed/suspected seizures.
- The combination of a proton pump inhibitor and clopidogrel has not been definitively shown to increase the risk for cardiovascular events; if the combination is used, omeprazole is recommended based on COGENT trial data.
- Guidelines strongly recommend therapy for speech-language pathology and partner training for aphasia.
- Poststroke, exercise is recommended for at least 10 minutes four times a week of moderate intensity or 20 minutes two times a week of vigorous intensity.

Introduction

For anyone practicing neurorehabilitation, it is important to understand stroke—from the recognition and acute treatment to postacute and chronic management. Stroke is one of the most common diagnoses in acute and subacute inpatient rehabilitation. Furthermore, secondary stroke is a common complication of stroke and may occur while a patient is still in the inpatient rehabilitation setting. Early recognition and management in this setting is critical. The mortality from stroke has decreased because of better education of the lay population in recognizing the symptoms, more aggressive management in the field, the availability of more effective treatments, and faster time to treatment. As a result, stroke went from the third leading cause of death in the United States as recently as 2008[54,65,113] to the fifth leading cause in 2016.[51,54] It remained the fifth leading cause in 2021 because COVID was third (2.5 times higher than stroke). Provisional data from the Centers for Disease Control and Prevention suggest the incidence has increased year after year between 2020 and 2023, and stroke is now the fourth leading cause of death. The overall prevalence of stroke is 3.1% among Americans over the age of 18 years. This number is more than double in those over age 65 years and increases to nearly 12% in those over age 75 years.[21,22,73,124] With the increased survival come increased numbers of people living with disability and increasing numbers of secondary strokes. One in four people who survive a stroke will have a recurrence within 5 years.[87] The decrease in mortality results in an overall increase in cost. This is in part because of the expense of some newer, more effective treatments as well as the long-term cost of survival. The cost of caring for stroke patients in the United States is estimated to be nearly $60,000 per person per year.[22,42,106] When you include indirect costs, such as lost wages, the total annual societal cost in the United States is approximately $103 billion.[42,92,93] This chapter focuses primarily on ischemic stroke because that accounts for more than 85% of all strokes.[22,51,54] However, it also discusses hemorrhagic strokes, arteriovenous malformations (AVMs), and aneurysm rupture.

Types of Strokes

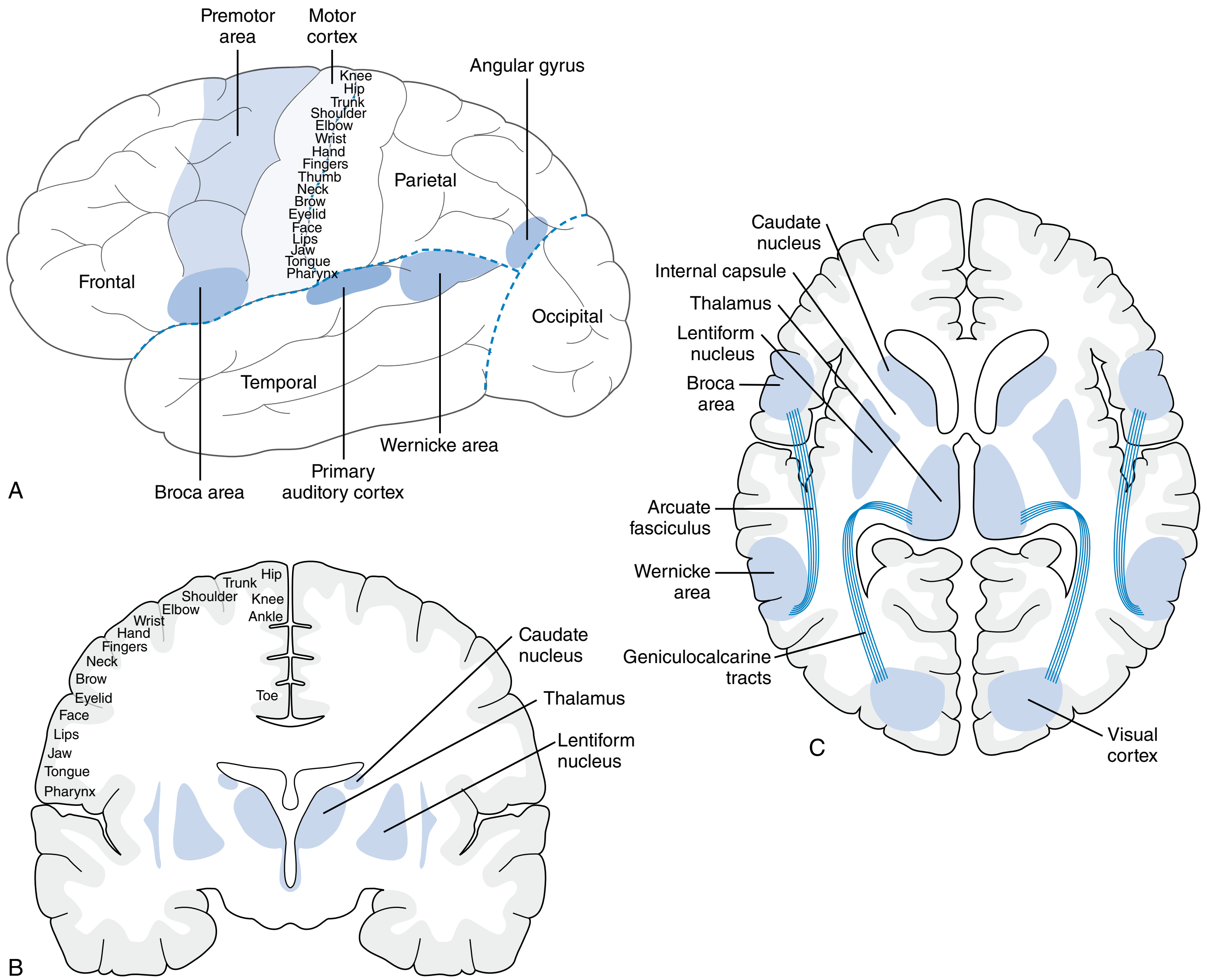
Strokes can be categorized in different ways. Ischemic strokes are the most common, accounting for approximately 85% to 90% of strokes.[36,51,73] In addition to nearly 800,000 strokes annually, there are 240,000 transient ischemic attacks (TIAs).[118] Risk factors include smoking, diabetes, hypertension, among other cardiovascular risk factors. Ischemic strokes are further divided by location or vascular territory. The most common type of ischemic stroke in the United States is a large-vessel thrombotic stroke in the territory of the middle cerebral artery (MCA). Large-vessel atherosclerotic stenosis and occlusion is less common in the United States than in other areas of the world; it accounts for 10% to 15% of strokes in the United States.[26,73,90] Of those, the majority are MCA strokes. Ischemic strokes from thrombosis or stenosis may occur in any large-vessel vascular territories.[45] Lacunar strokes are most often associated with hypertension.

Strokes seen at the border zone of two vascular territories ("border zone" or "watershed" infarct) are more frequently seen hypotension. They are commonly associated with cardiac surgeries caused by hypoperfusion that occurs during the procedure, or they may occur from microemboli seen from a cardiac etiology.[44]

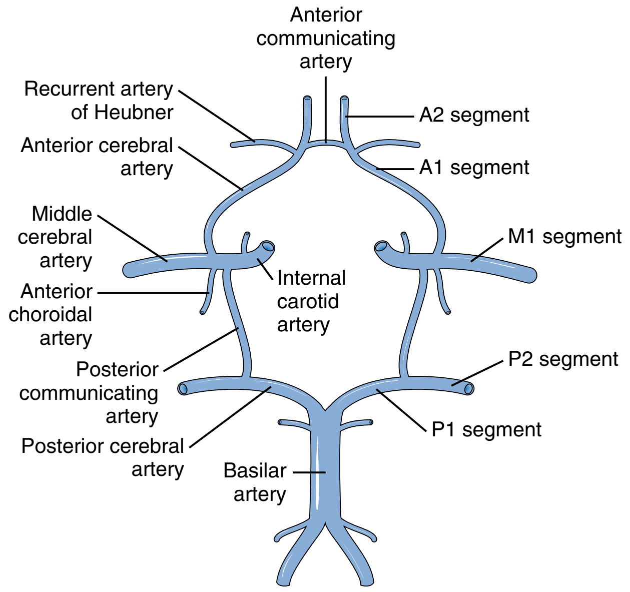
Intracerebral hemorrhage accounts for nearly 10% to 15% of all strokes, but with a much higher morbidity and mortality with only 27% of survivors being independent by 90 days poststroke and a mortality rate of 40% to 50%, which is twice that of ischemic stroke.[11,41] These hemorrhagic strokes occur from a variety of underlying causes, hypertension being one common cause. The most common areas for a hemorrhagic stroke caused by hypertension are the basal ganglia, thalamus, cerebellum, and pons.[103,123] Lobar hemorrhages from amyloid angiopathy are more often seen in those age 60 years and older. Some authors estimate that cerebral amyloid angiopathy accounts for up to 15% of intracerebral hemorrhages in this age group.[11,66,89,123] Although not considered "strokes," nontraumatic hemorrhages may also result from other causes. An AVM or aneurysm may rupture and cause an intracerebral hemorrhage. Subdural hematomas (SDHs) arise from the rupture of bridging veins because of their vulnerability to shear injury. Subarachnoid hemorrhages (SAHs) are located between the arachnoid and pia mater, tracking down the sulci and following the contours of the brain. The most common cause of SAH is rupture of a berry aneurysm; the most frequent locations for a ruptured aneurysm are the anterior cerebral artery (ACA) (27%–35%), internal carotid artery (ICA) (15%–20%), posterior communicating (PComm) artery (25%), MCA (8%–20%), and vertebrobasilar system (5%–15%).[8,10,82]

Basic Neuroanatomy: A Brief Overview

An understanding of basic neuroanatomy is important to help localize a stroke, determine management, and sometimes differentiate between a stroke and stroke mimics, such as migraine, seizure, and conversion disorder.[94] This section includes the anatomical location, blood supply (Figs. 45.1–45.3), and basic



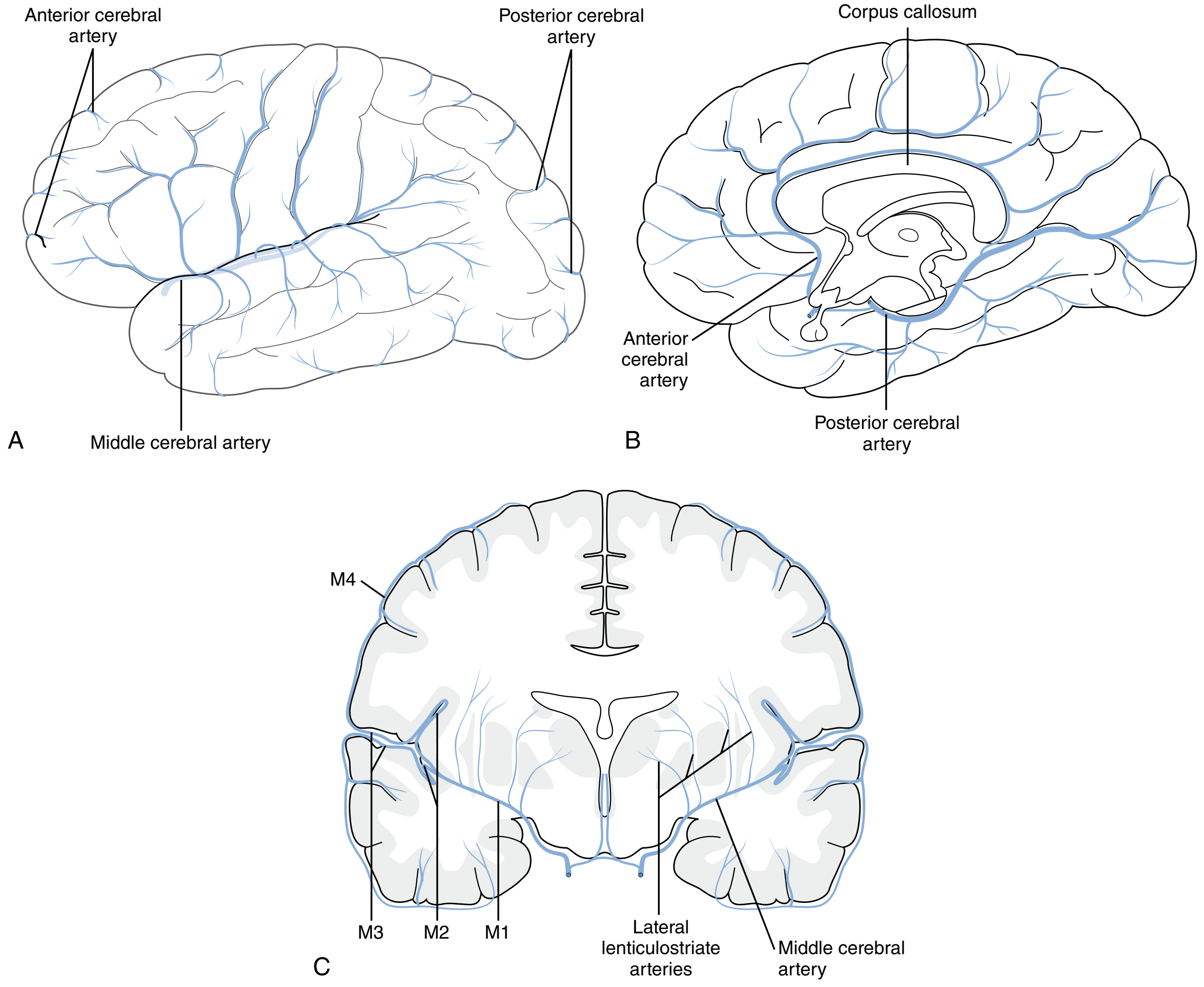
• **Fig. 45.1** (A–C) Surface anatomy of the cerebral cortex. (From https://netterimages.com/cerebrum-lateral-views-labeled-anatomy-atlas-5e-physiology-frank-h-netter-5274.html.)



• **Fig. 45.2** Territories of the cerebral arteries. (From https://www.netterimages.com/arteries-of-brain-frontal-view-and-section-labeled-anatomy-atlas-4e-general-anatomy-frank-h-netter-4588.html.)

function of different areas of the brain and brainstem (Tables 45.1–45.4). It is important to note that because of various pathways and circuits throughout the central nervous system (CNS), there is substantial overlap in function.

- The *frontal lobe* is the anterior portion of the brain; it extends to the central sulcus (or sulcus of Rolando) posteriorly and is inferiorly and laterally separated from the temporal lobes by the sylvian fissure. The gyrus immediately in front of the central sulcus is the precentral gyrus *(primary motor cortex)*. The olfactory bulbs and tract run along the inferior aspect of the frontal lobe bilaterally near the interhemispheric fissure. The most anterior portion is the frontal pole.
- The *parietal lobe* is separated from the frontal lobe by the central sulcus anteriorly but has no clear demarcation from the temporal lobes inferolaterally. The demarcation for the occipital lobe is at the parietooccipital sulcus. Directly posterior to the central sulcus lies the postcentral gyrus/sulcus *(primary somatosensory cortex)*.
- The *temporal lobe* has no clear demarcation from the parietal or occipital lobes; it lies inferior to the frontal lobe, separated by the sylvian fissure. The most anterior portion is the temporal pole. The *primary auditory cortex* is located near the sylvian fissure in the superior medial portion of the temporal lobe.



• **Fig. 45.3** (A–C) Circle of Willis. (From https://www.netterimages.com/cerebral-arterial-circle-willis-labeled-anatomy-atlas-5e-general-anatomy-frank-h-netter-4694.html.)

TABLE 45.1 **Cerebellar Stroke Syndromes**

Rostral cerebellar syndrome	Subthalamic region, thalamus, occipitotemporal lobes, lateral tegmental area of upper pons	Coma ± tetraplegia, ipsilateral dysmetria, Horner syndrome, contralateral pain/temp deficit, cranial nerve IV palsy, dysarthria, headache, dizziness, emesis, delayed coma
Medial cerebellar syndrome	Lateral area of the lower pons	Ipsilateral cranial nerve V, VII, VIII, Horner syndrome, dysmetria, contralateral pain/temp deficit
Caudal cerebellar syndrome	Dorsolateral medullary area	Vertigo, headache, emesis, ataxia, delayed coma

Modified from Zorowitz RD. Stroke syndromes: infratentorial. In: Stein J, Harvey RL, Winstein CJ, et al., eds. Stroke Recovery and Rehabilitation. 2nd ed. Demos Medical Publishing; 2015.

TABLE 45.2 **Major Artery Clinical Syndromes**

Artery	Deficits
L MCA-S	R face/arm weakness, Broca/nonfluent aphasia, ± sensory loss
L MCA-I	Fluent/Wernicke aphasia, R visual field deficit. R face/arm ± sensory loss; motor findings absent; confused; mild R weakness
L MCA-D	R pure motor hemiparesis, larger infarcts may produce cortical deficits
L MCA stem	Combination of all above with R hemiplegia/anesthesia, R homonymous hemianopia and global aphasia; L gaze preference
R MCA-S	L face/arm weakness, L hemineglect to a variable extent, ± sensory loss
R MCA-I	Profound L hemineglect; L visual field and somatosensory deficits but hard to test because of neglect; mild L weakness may be present; R gaze
R MCA-D	L pure motor hemiparesis, larger infarct can include cortical and neglect
R MCA stem	Combination of above with L hemi, L homonymous hemi, and profound L neglect with R gaze preference
L ACA	R leg weakness with sensory loss, grasp reflex, frontal lobe behaviors, transcortical aphasia ± larger may cause R hemi with lower extremity > upper extremity relatively
R ACA	L leg weakness with sensory loss, grasp reflex, frontal behaviors, L hemineglect; larger infarcts may cause full L hemi
L PCA	R homonymous hemianopia; extension to the splenium of the corpus callosum can cause alexia w/o agraphia; larger infarcts with thalamus and IC may cause aphasia, R hemisensory loss, and R hemiparesis
R PCA	L homonymous hemianopia; larger infarcts with thalamus and IC may cause L hemisensory loss and L hemiparesis

ACA, Anterior cerebral artery; *D,* deep; *I,* inferior; *IC,* internal capsule; *L,* left; *MCA,* middle cerebral artery; *PCA,* posterior cerebral artery; *R,* right; *S,* superior.
Modified from Blumenfeld H. Neuroanatomy Through Clinical Cases. 2nd ed. Sinauer Associates; 2010.

TABLE 45.3 **Common Lacunar Syndromes**

Syndrome	Features	Locations	Vessels
Pure motor hemiparesis or dysarthria hemiparesis	Contralateral face, arm, and leg weakness with dysarthria	Posterior limb IC, Corona radiata, Cerebral peduncle	Lenticulostriate, anterior choroidal or perforating branches of PCA, small MCA and basilar branches
Pure sensory stroke	Sensory loss—contralateral face and body	Thalamus, corona radiata, posterior limb IC	Thalamoperforator branches of the PCA
Sensorimotor stroke	Combination of thalamic and pure motor hemiparesis	Posterior limb IC and either VPL or thalamic sensory	Thalamoperforator branches of the PCA or lenticulostriate
Ataxic hemiparesis	Same as pure motor but with ataxia on same side	Posterior limb IC, corona radiata, cerebral peduncle, red nucleus, lentiform nucleus	Same as pure motor hemiparesis
Basal ganglia stroke	Usually asymptomatic, but may cause hemiballismus	Caudate, putamen, globus pallidus, or subthalamic nucleus	Lenticulostriate, anterior choroidal, thalamoperforator, or Heubner arteries

IC, Internal capsule; *MCA,* middle cerebral artery; *PCA,* posterior cerebral artery; *VPL,* ventral posterolateral nucleus.
Modified from Blumenfeld H. Neuroanatomy Through Clinical Cases. 2nd ed. Sinauer Associates; 2010.

TABLE 45.4 Common Brainstem Syndromes

Region	Name	Supply	Structures	Features
Medulla				
Medial medulla	Medial medullary syndrome	Branch of vertebral and anterior spinal	Pyramidal tract and medial lemniscus	Contralateral arm/leg weakness, decreased position and vibration
Lateral medulla	Wallenberg syndrome	Vertebral > PICA	CN XII, vestibular, and trigeminal nuclei, cerebellar peduncle, spinothalamic tract, sympathetic tract, nucleus ambiguus and solitarius	Ipsilateral ataxia, vertigo, nystagmus, nausea, face decreased pain and temperature, Horner syndrome and decreased taste. Contralateral body decreased pain/temp, hoarseness, dysphagia
Pons				
Medial pons basis	1. Dysarthric hemiparesis 2. Ataxic hemiparesis	Paramedian branches of the basilar and vertebral	1. Corticospinal/corticobulbar tract 2. Above + pons nuclei and pontocerebellar tract	1. Contralateral face/arm/leg weakness, dysarthria 2. All the above plus contralateral ataxia
Medial pons basis/ tegmentum	1. Millard-Gubler syndrome 2. Foville syndrome	Branches of basilar, ventral, and dorsal territories	1. Corticospinal/corticobulbar tract 2. Corticospinal/corticobulbar tract plus facial colliculus	1. Contralateral face/arm/leg weakness 2. Contralateral face/arm/leg weakness plus dysarthria with ipsilateral face weakness/gaze
Lateral caudal pons	AICA syndrome	AICA	Middle cerebellar peduncle, vestibular, trigeminal nuclei, spinothalamic, and sympathetic tracts	Ipsilateral ataxia, facial decrease in pain/temp, Horner syndrome, contralateral body decrease in pain/temp, vertigo, nystagmus
Dorsolateral rostral pons	SCA syndrome	SCA	Superior cerebellar peduncle, lateral structures	Ipsilateral ataxia plus possible AICA inclusions
Bilateral pons	Locked-in syndrome	Basilar artery	Bilateral corticospinal/corticobul-bar/pontine, reticular formation	Tetraplegia, facial palsy with sparing of eye blinking, bilateral horizontal gaze palsy
Midbrain				
Midbrain basis	Weber syndrome	PCA, basilar	CN III, cerebral peduncle	Ipsilateral third nerve palsy and contralateral hemiparesis
Midbrain tegmentum	Claude syndrome	PCA, basilar	CN III, superior cerebral peduncle, red nucleus	Ipsilateral third nerve palsy, contra-lateral ataxia, rubral tremor
Midbrain basis and tegmentum	Benedikt syndrome	PCA, basilar	All of the above plus substantia nigra ± ML and ST tract	Ipsilateral third nerve palsy, contra-lateral hemiparesis, ataxia, tremor and involuntary movements
Dorsal rostral midbrain	Parinaud syndrome	PCA, basilar	Rostral interstitial MLF	Multiple oculomotor abnormalities include upgaze palsy, tonic downward gaze ("setting sun"), vertical nystagmus, bilateral ptosis
Rostral midbrain	Top of the basilar syndrome	PCA, basilar	Rostral midbrain, thalami, posterior temporal and occipital lobes	Parinaud + abulia, hypersomno-lence, memory impairment

AICA, Anteroinferior cerebellar artery; *ML,* medial lemniscus; *MLF,* medial longitudinal fasciculus; *PCA,* posterior cerebral artery; *PICA,* posteroinferior cerebellar artery; *SCA,* superior cerebellar artery; *ST,* spinothalamic tract.

Modified from Blumenfeld H. Neuroanatomy Through Clinical Cases. 2nd ed. Sinauer Associates; 2010.

- The *occipital lobe* has no clear demarcation from the temporal lobe. The parietooccipital sulcus separates it from the parietal lobe. The most posterior portion is the occipital pole. Directly inferior to the occipital lobe is the cerebellum. The *primary visual cortex* is located in the occipital lobe.
- The *insular cortex* is covered laterally by a lip of the frontal cortex anteriorly and parietal cortex posteriorly.
- *Subcortical* refers to several areas, including the *caudate nucleus* and *putamen (striatum)*, *globus pallidus, subthalamic nucleus*, and *substantia nigra.* The caudate and putamen are separated by the

internal capsule. The putamen forms the lateral portion, fusing with the head of the caudate anteriorly and ventrally. Just medial to the putamen is the globus pallidus, which has internal and external segments. The thalamus sits nearby, separated from the *lentiform nucleus* by the posterior limb of the internal capsule. The substantia nigra is just dorsal to the *cerebral peduncles* and is separated from the globus pallidus by the internal capsule.

- The *brainstem* is divided into the midbrain, pons, and medulla. The *midbrain* connects the pons and cerebellum to the thalamus and cerebral hemispheres. It contains the cerebral peduncles as well as the aqueduct between the third and fourth ventricles. The *pons* links information from the medulla to the higher cortical areas of the cortex and is also connected to the cerebellum through the middle cerebellar peduncle; it includes the pontine reticular formation as well as centers for respiration. The *medulla* contains the crossing of tracts relaying information between the cortex and spinal cord. It contains centers for respiration as well as vasomotor and cardiac function and the mechanisms for cough, gag, and swallowing.
- The *cerebellum* consists of a midline vermis and two cerebellar hemispheres. The largest fissure is called the primary fissure and separates the cerebellum into anterior and posterior lobes. The posterolateral fissure separates the posterior lobe from the flocculonodular lobe. The tonsils are at midline, and laterally there are three cerebellar peduncles: the superior (which is the most medial), the middle (which is the most lateral), and the inferior (which is the most inferior).[14,44,126]

Vascular Anatomy

The circle of Willis (see Fig. 45.2) is supplied by the right and left ICAs anteriorly and the basilar artery posteriorly. Before the bifurcation of the MCA and ACA, the ICA gives off the ophthalmic, anterior choroidal, and PComm arteries. From the ACA arise the anterior communicating artery, recurrent artery of Heubner, and the pericallosal and medial lenticulostriate arteries. The MCA provides the lateral lenticulostriate, branching into the superior (S), inferior (I), and deep (D) branches bilaterally (MCA-S, MCA-I, and MCA-D). The basilar artery bifurcates into the posterior cerebral artery (PCA), with the circle being completed by the anastomoses of the PComm arteries with the PCA bilaterally. The major vascular anatomy of the brain is shown in Fig. 45.3. The vertebral arteries provide the majority of the blood supply to the brainstem structures and merge into the basilar artery, which provides the pontine and medullary branches. The posteroinferior cerebellar artery (PICA) supplies the medulla as well as portions of the spinothalamic tract. The anteroinferior cerebellar artery (AICA) supplies the pons, the facial and spinal trigeminal nucleus and tract, as well as the inferior and middle cerebellar peduncles and spinothalamic tract. Blood supply for the cerebellum is provided by three branches of the vertebral and basilar arteries. The PICA supplies the lateral medulla, inferior half of the cerebellum, and inferior vermis. The AICA supplies the inferolateral pons, middle cerebellar peduncle, and ventral anterior strip of cerebellum between the PICA and superior cerebellar artery (SCA) territories. The SCA supplies the upper lateral pons, superior cerebellar peduncle, and most of the superior half of the cerebellar hemisphere.[14,126]

Initial Workup/Management

This section refers primarily to those patients being evaluated for a first stroke (or a recurrence >3 months after a previous stroke). Many of the principles in this section apply even during a recurrence in the first 30 days, which is the period during which the patient is at highest risk for recurrence. However, certain contraindications, particularly related to anticoagulation, in the setting of acute stroke are also discussed.

Just as the general population should be able to recognize the warning signs of stroke, so should the nurses and other clinicians caring for patients who have had strokes. Symptoms include the acute onset of focal neurological symptoms (e.g., weakness, dizziness, or slurred speech), mental status changes, or sudden severe headaches. Initiation of an emergency response process is paramount because time is of the essence. Standard protocols—including assessment, obtaining intravenous access, administration of oxygen and appropriate fluids, and management of hypotension and hypoglycemia—should be followed. The emergency department should be ready to receive a stroke patient and have the physician assessment, including the National Institutes of Health Stroke Scale (NIHSS), completed within minutes of arrival, and a noncontrast computed tomography (CT) image of the head within 20 minutes of arrival. A noncontrast magnetic resonance imaging (MRI) scan of the brain using diffusion-weighted imaging (DWI) is more sensitive than CT to ischemia, particularly in the posterior fossa and brainstem, but it is often not feasible because of access or time constraints.[57] Evaluation should include a thorough history to exclude stroke mimics (e.g., seizure, migraine, conversion disorder). These mimics must be considered in patients who are being evaluated for recurrent strokes as well as for primary stroke.[94] Seizures should be considered if the new neurological deficit correlates with the same territory as the initial stroke. If it is determined that the patient has sustained an ischemic stroke and completes the workup within 4.5 hours of onset, the patient is a candidate for intravenous recombinant tissue plasminogen activator (rtPA) alteplase.[29,57] There is very clear decrement in benefit and increased risk of complications as time from onset elapses. The benefit of rtPA clearly outweighs the risk in the first 3 hours. The benefit-risk ratio worsens over the next 90 minutes, but the risk for severe complications, primarily intracranial hemorrhage, still remains relatively low. Between 4.5 and 6 hours, however, the risk of intracranial hemorrhage increases, and the benefit decreases to the point that rtPA is no longer recommended beyond the 4.5-hour window. Of note, there are multiple contraindications to the administration of rtPA, one of which is prior stroke or serious head trauma within the preceding 3 months. Therefore this would rarely be appropriate for a patient who experiences a recurrent stroke during the inpatient rehabilitation period.[39] A cardiac evaluation should include an electrocardiogram, troponin, and chest x-ray; however, these test results should not delay administration of rtPA. Unlike intravenous rtPA, intraarterial thrombolytics can be given up to 6 hours after stroke onset. Additional evaluation should include the following laboratory studies: complete blood count, prothrombin time, partial thromboplastin time, glucose, hemoglobin A1c, creatinine, and a lipid panel (including low-density lipoprotein [LDL] cholesterol).[64]

In the inpatient rehabilitation setting, a stroke patient with a new or worsening neurological deficit should be evaluated with noncontrast CT to ensure there has not been hemorrhagic conversion. Given the increased risk of seizures poststroke (~10% incidence),[12,46] an electroencephalogram (EEG) is indicated if the neuroimaging does not indicate a new stroke or bleed.

Mechanical thrombolysis expands the treatment window of acute stroke up to 24 hours.[24,43,100,107] However, it is limited to large-vessel thrombotic strokes typically affecting the carotid or MCA territory. These techniques can also be used in patients who have already received intravenous rtPA.[89] It is possible that these interventions could also be used in someone who has had a stroke more recently; however, the studies typically required a prestroke Rankin score of 0 to 1, so it remains unclear whether it would be appropriate for patients who are still in inpatient rehabilitation because they would have a higher Rankin score. The DAWN and DEFUSE 3 studies demonstrate benefit of mechanical thrombolysis up to 24 and 16 hours, respectively, after stroke onset, allowing these treatments to be used for "wakeup" strokes.[5,89] Previously, treatment in this population was extremely limited because, if clinicians were unable to determine the exact time for onset of symptoms, the patient would be ineligible for treatment with intravenous rtPA and usually also ineligible for intraarterial thrombolysis.

In the acute period, hypertension is more commonly seen than hypotension. "Permissive hypertension" is reasonable, but significant elevations, particularly if a fibrinolytic therapy is being used, should be corrected. Overcorrection must be avoided because arterial hypotension is associated with a worse outcome. If the patient received fibrinolytic therapy, blood pressure should be maintained below 185 mm Hg systolic and 110 mm Hg diastolic. If no fibrinolytic therapy was used, the blood pressure may be allowed to go to 220 mm Hg systolic and 120 mm Hg diastolic. For patients with hypertensive lacunar infarcts and those with diabetes, it is recommended that their systolic blood pressure be maintained below 140 mm Hg. If the blood pressure must be managed, an appropriate goal is a decrease of 15% gradually over 24 hours.[11,63,64,106] In addition to blood pressure control, the patient's oxygen saturation should be maintained at 94% or higher. Hypovolemia should be rapidly treated, and maintenance intravenous fluids should be run.

Hyperglycemia should also be treated because it has been associated with worse outcomes. Very tight control is inadvisable because the patient could become hypoglycemic, particularly in the postacute period when the patient may have a lower caloric intake than usual and is increasing physical activity during therapy. A goal of 140 to 180 mg/dL in the acute period is reasonable.[57]

There are some distinct differences in the management of ischemic stroke vs. spontaneous intracranial hemorrhage (hemorrhagic stroke). Unlike in ischemic stroke, in the case of intracranial hemorrhage, blood pressure should be rapidly reduced to a goal of systolic blood pressure 130 to 150 mm Hg within 1 hour of onset with a long-term goal of less than 130/80 mm Hg. If the patient is on anticoagulation, the agent should immediately be discontinued and reversal agents administered. For patients on antiplatelet agents, platelet transfusions should only be considered if the patient is undergoing emergency neurosurgery. Prophylactic treatment for seizures is not indicated, but treatment should be initiated in patients with EEG confirmed or clinically observed/suspected seizures. Early mobilization within the first 24 hours of intracranial hemorrhage may increase mortality; however, it is safe to begin mobilization after 24 to 48 hours.[49]

Anticoagulation is often necessary for the treatment of venothrombosis, atrial fibrillation, or other cardiac issues. It is important not to start those agents within 24 hours of the use of intravenous rtPA. There is no indication for emergent anticoagulation in the face of acute stroke or neurological worsening. The risk of intracranial hemorrhage is increased, and an improved outcome has not been demonstrated. It has not been shown to reduce the risk of stroke recurrence even in the face of a cardioembolic stroke.[57] Dabigatran has been approved for the prevention of stroke associated with atrial fibrillation, but the optimal timing of its initiation poststroke has not been established.

Antiplatelet agents should be used to reduce the risk of stroke and stroke recurrence.[26] Although the specific guidelines for use of single antiplatelet therapy vs. dual antiplatelet therapy (DAPT) may vary depending on the specific underlying cause of stroke, for noncardioembolic ischemic stroke (the most common cause) or TIA, DAPT is recommended for a mild stroke (NIHSS ≤ 3) or after TIA in a high-risk patient ($ABCD^2$ score ≥ 4). In most other situations, the benefit of DAPT compared with single agent therapy has not been demonstrated, and the risk of bleeding increases. For cases in which DAPT is indicated, it should be initiated within 24 to 48 hours of the onset of symptoms, and it should continue for 21 to 90 days at which time it should be changed to single antiplatelet treatment. Currently, there are no clear guidelines on the best antiplatelet regimen for a patient who has had a stroke while taking aspirin.[51,64]

There is often a question of when to restart anticoagulation after a hemorrhagic conversion of an ischemic stroke. This may be for prevention of thromboembolism or, in the case of patients with atrial fibrillation, recurrence of ischemic stroke. Although there is variability in the literature, in most cases restarting after 14 days posthemorrhage is probably a reasonable guideline. In cases of very large hemorrhages, particularly ones resulting in increased intracranial pressure and/or requiring a decompressive craniectomy, it would be prudent to wait 4 weeks. The data suggest that starting early (2 weeks) does not increase the mortality nor the risk of intracranial hemorrhage in most cases. The risk of intracranial hemorrhage from anticoagulation is lower after traumatic hemorrhages than from a hemorrhagic conversion of stroke; therefore the earlier resumption is reasonable in patients who are at high risk for stroke.[49,69]

There is a great deal of interest in neuroprotective agents, several of which have shown efficacy in animal studies. To date there is very limited evidence of effective neuroprotection in human studies. This may be caused by differences between the study animals and humans or, in some cases, may just be a timing issue. In the case of animal studies, the timing of administration of a neuroprotective agent can be hyperacute, whereas in human studies a delay in administration may render a treatment ineffective. Intravenous magnesium has been studied and has not shown benefit when given within 12 hours with the possible exception of a moderate effect in lacunar strokes.[83,91] It is also likely that to be effective, neuroprotective treatment will need to target multiple pathways simultaneously. Current pathways that are most promising include drugs that target inflammatory and neuroexcitatory pathways.[94] Statins have been felt to have a neuroprotective benefit because of their direct influence on endothelial function, reduction of oxidative stress, antiinflammatory properties, influence on plaque stability, and other properties. Although neuroprotection has not been clearly established, patients taking a statin at the time of their stroke should continue this because, in a small randomized trial, it was noted that those who discontinued had greater odds of death or dependency at 3 months poststroke.[13,57] Although this does not necessarily imply that starting a patient on a statin poststroke will provide neuroprotection, it is recommended to continue the statin in those patients who had already been taking it.

It is estimated that 90% of strokes result from modifiable causes, thus the importance of educating patients and ensuring appropriate secondary stroke measures are in place after a patient has had a stroke. In addition to appropriate antiplatelet or anticoagulation and a statin, recommendations may include regular aerobic exercise, weight loss, a Mediterranean low-sodium diet, and reduction in alcohol.[36,64]

Complications of Stroke

Guidelines for the management of adult stroke rehabilitation and recovery have been published by the American Heart Association (AHA) and the American Stroke Association (ASA) and have been endorsed by the American Academy of Physical Medicine and Rehabilitation and the American Society of Neurorehabilitation.[49,64,122] These guidelines will be referenced in the discussion of both the prevention and management of complications and the rehabilitation treatment of stroke in this chapter, extensively referring to levels of evidence (Fig. 45.4). In the majority of cases, references to levels of evidence using the Level A to C/Class I to III system are in reference to the 2016 AHA/ASA Guidelines for Adult Stroke Rehabilitation and Recovery. Additionally, the 2021 AHA/ASA Guidelines on Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack[64] are referenced in relation to issues of secondary stroke prevention related to cardiometabolic risk factors and substance abuse.

To provide excellent stroke rehabilitation, it is critical for the rehabilitationist to prevent, identify, and correctly treat the common complications that follow stroke. Overall, complications are frequent and are seen in up to 85% of stroke patients during hospitalization (Table 45.5).[67]

An autopsy study[101] looking at cause of death in a sample of 409 stroke patients identified different likely diagnoses at different times out from stroke. The study evaluated 82 of the 95 patients who died within the first 3 months. Direct effects of stroke were the most common cause of death in the first week (90%), pulmonary embolism (PE) most common in the second to fourth weeks (30%), and bronchopneumonia in the second to third months (27%). Of the 61 of 128 patients who died after 3 months, cardiac disease was the most common cause (37%). Overall, death was due to causes other than stroke in 59% of ischemic stroke patients and 24% of hemorrhagic stroke patients. These data point to the importance of effective and mindful treatment in the rehabilitation environment in relation to venous

SIZE OF TREATMENT EFFECT

ESTIMATE OF CERTAINTY (PRECISION) OF TREATMENT EFFECT

	CLASS I	**CLASS IIa**	**CLASS IIb**	**CLASS III** *No Benefit* or **CLASS III** *Harm*
	Benefit >>> Risk Procedure/Treatment **SHOULD** be performed/ administered	*Benefit >> Risk* *Additional studies with focused objectives needed* IT IS REASONABLE to perform procedure/administer treatment	*Benefit ≥ Risk* *Additional studies with broad objectives needed; additional registry data would be helpful* Procedure/Treatment MAY BE CONSIDERED	Procedure/Test — Treatment COR III: No benefit: Not Helpful — No Proven Benefit COR III: Harm: Excess Cost w/o Benefit or Harmful — Harmful to Patients
LEVEL A Multiple populations evaluated* Data derived from multiple randomized clinical trials or meta-analyses	• Recommendation that procedure or treatment is useful/effective • Sufficient evidence from multiple randomized trials or meta-analyses	• Recommendation in favor of treatment or procedure being useful/effective • Some conflicting evidence from multiple randomized trials or meta-analyses	• Recommendation's usefulness/efficacy less well established • Greater conflicting evidence from multiple randomized trials or meta-analyses	• Recommendation that procedure or treatment is not useful/effective and may be harmful • Sufficient evidence from multiple randomized trials or meta-analyses
LEVEL B Limited populations evaluated* Data derived from a single randomized trial or nonrandomized studies	• Recommendation that procedure or treatment is useful/effective • Evidence from single randomized trial or nonrandomized studies	• Recommendation in favor of treatment or procedure being useful/effective • Some conflicting evidence from single randomized trial or nonrandomized studies	• Recommendation's usefulness/efficacy less well established • Greater conflicting evidence from single randomized trial or nonrandomized studies	• Recommendation that procedure or treatment is not useful/effective and may be harmful • Evidence from single randomized trial or nonrandomized studies
LEVEL C Very limited populations evaluated* Only consensus opinion of experts, case studies, or standard of care	• Recommendation that procedure or treatment is useful/effective • Only expert opinion, case studies, or standard of care	• Recommendation in favor of treatment or procedure being useful/effective • Only diverging expert opinion, case studies, or standard of care	• Recommendation's usefulness/efficacy less well established • Only diverging expert opinion, case studies, or standard of care	• Recommendation that procedure or treatment is not useful/effective and may be harmful • Only expert opinion, case studies, or standard of care
Suggested phrases for writing recommendations	Should is recommended is indicated is useful/effective/ beneficial	is reasonable can be useful/effective/ beneficial is probably recommended or indicated	may/might be considered may/might be reasonable usefulness/effectiveness is unknown/unclear/uncertain or not well established	COR III: No Benefit: is not recommended; is not indicated; should not be performed/ administered/ other; is not useful/ beneficial/ effective COR III: Harm: potentially harmful; causes harm associated with excess morbidity/ mortality; should not be performed/ administered/ other
Comparative effectiveness phrases[†]	treatment/strategy A is recommended/indicated in preference to treatment B treatment A should be chosen over treatment B	treatment/strategy A is probably recommended/indicated in preference to treatment B it is reasonable to choose treatment A over treatment B		

• **Fig. 45.4** Classification of recommendations and level of evidence. (Reprinted with permission. Stroke.2016;47:e98-e169. ©2016 American Heart Association, Inc.)

TABLE 45.5 Medical Complications After Stroke

Complication	Frequency
Recurrent stroke	9%
Epileptic seizure	3%
Urinary tract infection	23%
Chest infection	22%
Other infection	19%
Falls	26%
Falls with injury	5%
Pressure sores	21%
Deep vein thrombosis	2%
Pulmonary embolism	1%
Shoulder pain	9%
Other pain	34%
Depression	16%
Anxiety	14%
Emotionalism (pseudobulbar affect)	12%
Confusion	36%

Modified from Langhorne P, Stott DJ, Robertson L, et al. Medical complications after stroke: a multicenter study. Stroke. 2000;31(6):1223-1229.

thromboembolism (VTE), prevention of pulmonary complications, and appropriate medications and lifestyle adjustments to decrease the rate of cardiac disease after discharge.

Neurological Complications

Hemorrhagic Conversion

Hemorrhagic conversion is relatively common after ischemic stroke even when current acute stroke interventions, such as fibrinolysis or mechanical thrombectomy, have not been received; previous studies have noted an incidence of 40.6% after cerebral embolism[92] and 43% in cerebral infarction.[50] Okada et al. noted that bleeding risk was much higher in moderate- to large-size infarcts (50%) compared with small infarcts (2.9%) and that bleeding risk increased with age. However, only the more severe parenchymal hematoma type 2 (PH2) within the Fiorelli classification (Table 45.6)[37] was shown to be a significant predictor of neurological deterioration and 3-month mortality. The PH2 classification includes hemorrhagic conversion with a homogeneous hyperdensity occupying more than 30% of the infarct zone, significant mass effect, or any homogenous hyperdensity located beyond the borders of the infarct zone. The presence of either hemorrhagic infarction type 1 (HI1), hemorrhagic infarction type 2 (HI2), or parenchymal hematoma type 1 (PH1) was not shown to be associated with early neurological decline or 3-month mortality. The incidence of hemorrhagic transformation after intervention for acute stroke is complicated and beyond the scope of this chapter.[109]

TABLE 45.6 Fiorelli Classification of Intracranial Hemorrhage

Hemorrhage Classification	Radiographic Appearance
Hemorrhage infarction type 1	Small hyperdense petechiae
Hemorrhage infarction type 2	More confluent hyperdensity throughout the infarct zone; without mass effect
Parenchymal hematoma type 1	Homogeneous hyperdensity occupying <30% of the infarct zone; some mass effect
Parenchymal hematoma type 2	Homogeneous hyperdensity occupying >30% of the infarct zone; significant mass effect; or any homogenous hyperdensity located beyond the borders of the infarct zone

Modified from Fiorelli M, Bastianello S, von Kummer R, et al. Hemorrhagic transformation within 36 hours of a cerebral infarct: relationships with early clinical deterioration and 3-month outcome in the European Cooperative Acute Stroke Study I (ECASS I) cohort. Stroke. 1999;30(11):2280-2284.

Repeat Stroke

Because of more aggressive medical treatment specifically to reduce secondary strokes, the rate of recurrence of stroke after ischemic stroke and/or TIAs has dropped to about 3% to 4% per year.[78] The rate of recurrence is highest in the first 30 days after the initial stroke/TIA. Approximately one-half of all strokes that recur within 90 days occur within the first 2 weeks, thus highlighting the importance of aggressive measures and close monitoring during the acute inpatient rehabilitation period. The most important risk factor to target for secondary stroke prevention is hypertension.[30,63] In most cases, antihypertensives should be restarted 24 hours poststroke in any patient who was taking them before the stroke.[63] Once the period of permissive hypertension is passed, the 2021 AHA/ASA Guidelines on Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack recommend thiazide diuretics, angiotensin-converting enzyme inhibitors, or angiotensin II receptor blockers as useful options for lowering blood pressure (Class IA) to a recommended goal of less than 130/80 mm Hg for most patients (IB).[64]

Erdur et al.[34] in their study of 5106 patients reported on the risk of stroke recurrence during the period of acute inpatient hospitalization. The overall risk in their study was less than 1%, but the patients at highest risk for recurrence were those with a history of TIA, high-grade symptomatic carotid stenosis, or stroke secondary to another determined etiology (e.g., giant cell arteritis, cervical arterial dissection). Pneumonia poststroke was also associated with an increased risk of stroke recurrence, and aphasia was associated with a decreased risk. There was no difference in the medical management of those with and without recurrence.[34] When a recurrence of stroke is suspected during an inpatient rehabilitation stay, the patient should be evaluated by CT to rule out hemorrhagic transformation and ensure no acute neurosurgical intervention or management of intracranial hypertension is required. Assuming that that is not the case, evaluation with MRI, including DWI, will help to confirm a new stroke. Treatment for an acute stroke recurrence within the first 30 to 90 days follows the same guidelines as treatment for any other acute stroke with the exception of certain contraindications for treatment with thrombolytics or interventional procedures, as noted previously. If no new stroke has occurred, stroke mimics must be considered. Testing to evaluate for infection, most commonly urinary tract infection (UTI) or pneumonia, should be completed because infection can cause a worsening of neurological symptoms. Seizures must also be considered in the differential diagnosis as a stroke mimic, keeping in mind that nonconvulsive status epilepticus (NCSE) can present as mental status changes and may be quite subtle. In one study of 889 stroke patients, NCSE was found in 3.6%.[9,79]

Seizures

The incidence of seizure after a stroke varies widely in the literature depending on the study. In the early (first week) poststroke period, it is estimated to be 2% to 23%; this increases to up to 3% to 67% chronically. Based on our clinical experience and some of the cited literature,[4,12] it is more likely that the actual incidence is on the lower end of the range. Of use to the rehabilitationist, one survey of a tertiary stroke rehabilitation program identified an incidence of only 1.5%, and Langhorne et al. noted a 3% incidence during hospitalization after stroke and an eventual 5% incidence with follow-up up to 30 months.[67,75] Should a seizure occur, a workup for contributing causes should be pursued and the addition of antiepileptic medications should be considered (IC, e13). The guidelines suggest "routine prophylaxis for patients with ischemic or hemorrhagic stroke is not recommended" (IIIC, e13).[122]

AHA/ASA guidelines for intracerebral hemorrhage note a risk of early seizures up to 16%, with the majority of these occurring at onset. Risk of seizure is increased with cortical involvement. Additionally, on continuous EEG monitoring, up to 28% to 31% of a group of individuals with intracerebral hemorrhage were found to have evidence of seizures despite the fact that most were receiving prophylaxis. Ultimately, given a lack of difference in outcomes, the guidelines note an absence of convincing evidence to recommend the prophylactic use of anticonvulsants; however, they do recommend treating identified clinical or electrographic seizures.[53] The AHA/ASA guidelines for SAH management note a likely 3% to 7% risk of late seizures, although there is no consensus on either the immediate or chronic use of anticonvulsants. The risk of late seizure may be increased among those with a thicker clot or rebleeding, and it is left to the discretion of the provider whether to administer prophylaxis, balancing the negative effects of anticonvulsants with the presumed risk of seizure.[27,82]

Hydrocephalus

Acute hydrocephalus is common after aneurysmal SAH, occurring in up to 15% to 87% of patients, whereas chronic, shunt-dependent hydrocephalus can be seen in 8.9% to 48% of patients.[27] Although these patients are often treated with an extraventricular drain, and hydrocephalus may apparently resolve in the acute stage, it is important for the rehabilitationist to recognize the risk of chronic hydrocephalus and to identify patients with a gradual decline in mental and functional status by implementing appropriate imaging and neurosurgical consultation for shunt placement as needed.

Cerebral Vasospasm

Though not frequently seen in the rehabilitation environment, large-artery narrowing after aneurysmal SAH is visible in up to 50% of cases, usually occurring 7 to 10 days after the aneurysm's rupture and resolving spontaneously after 21 days. Vasospasm may contribute to a secondary ischemic injury and lead to further morbidity and mortality. Guidelines suggest oral nimodipine (IA) and euvolemia with normal circulating blood volume (IB) for all patients with aneurysmal SAH to decrease the risk of vasospasm. Transcranial Doppler and perfusion imaging with CT may be used for monitoring.[26,27]

Venous Thromboembolism

In terms of VTE prevention, a process on admission should be implemented to ensure appropriate prophylaxis. Given the still notable risk for deep venous thrombosis (DVT) and PE in immobile patients, even in those appropriately treated with pharmacological or mechanical prophylaxis, a low threshold for repeat assessment for DVT and PE should be maintained even in the situation of appropriate prophylaxis. Illustrating this, Samama et al. noted a relative risk reduction of 63% with low-molecular-weight heparin (LMWH) compared with placebo in a population of acutely ill medical patients. This study demonstrated a DVT rate of 14.9% in individuals without prophylaxis and a rate of 5.5% in those on LMWH.[102] Guidelines support the use of prophylactic-dose heparin products during acute and rehabilitation hospitalizations up to the time that a patient regains mobility (IA). It is common practice to use ambulation of 150 ft as the threshold to discontinue prophylactic anticoagulation. After intracerebral hemorrhage, prophylactic-dose heparin products are recommended to begin 2 to 4 days in most cases (IIbC). In both ischemic stroke and intracerebral hemorrhage, LMWH is generally preferred over unfractionated heparin (IIaA and IIbC, respectively). If anticoagulation is contraindicated, intermittent pneumatic compression is preferable to no prophylaxis (IIbB and IIbC, respectively). In both ischemic stroke and intracerebral hemorrhage, elastic compression stockings are not recommended (IIIB and IIIC, respectively).[122]

Pulmonary Complications

Additional care should be taken in the identification of dysphagia and aspiration because the presence of these findings increases the risk of pneumonia by a relative risk of 3.17 and 11.56, respectively.[74] See Rehabilitation: Treatment of Stroke Sequelae (later) for recommendations on advancement to an oral diet.

Cardiovascular/Metabolic Complications

The AHA/ASA 2021 guidelines should be implemented with initiation of antiplatelet medications, anticoagulants, statins, and other risk-modifying medications as clinically indicated. A robust description of options is beyond the scope of this chapter. The 2021 AHA/ASA Guidelines on Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack suggest individuals "follow a Mediterranean-type diet with emphasis on monounsaturated fat, plant-based foods, and fish consumption, with either high extra virgin olive oil or nut supplementation, in preference to a low-fat diet, to reduce risk of recurrent stroke" and recommend reducing sodium intake by 1 g/day (IIaB).[64]

The 2021 AHA/ASA guidelines also address physical activity: recommending at least either 10 minutes four times a week of moderate intensity or 20 minutes twice a week of vigorous intensity (IC) as well as participation in a class that includes counseling on behavior change (IIaB), supervision by a health care professional (IIaC), and breaking up sedentary time with 3 minutes of standing or light exercise every 30 minutes (IIbB).[64]

In terms of hyperlipidemia, the 2021 guidelines recommend atorvastatin 80 mg for those with stroke with LDL cholesterol above 100 mg/dL with no known coronary heart disease or major sources of cardiac embolism. For those with atherosclerotic disease, a statin is recommended along with ezetimibe if needed to achieve a goal LDL cholesterol less than 70 mg/dL (IA). Fasting lipids should be checked 4 to 12 weeks after initiation of statin to determine efficacy and adjust the plan as needed (IA). Given this time frame, the rehabilitationist will be in the position to initiate further changes in medication management to minimize secondary stroke risk and should consider implementing measures to facilitate repeat testing. Should combination statin and ezetimibe therapy not be successful in lowering LDL cholesterol below 70, then PCSK9 inhibitors may be considered (IIaB). Though potentially useful in identifying the etiology of early atherosclerotic disease, guidelines do not currently provide guidance on assessing lipoprotein(a) (aka lipoprotein "little a") as an independent risk factor. Finally, icosapent ethyl (2 g twice daily) is a reasonable option for patients with stroke or TIA with fasting triglycerides 135 to 499 mg/dL with an LDL cholesterol of 41 to 100 mg/dL on moderate- to high-intensity statin with a hemoglobin A1c less than 10% and no history of pancreatitis or heart failure (IIaB).[64]

AHA/ASA 2021 guidelines recommend those with ischemic stroke or TIA should have a goal hemoglobin A1c of 7% or less for most patients (IA) and that those with diabetes receive treatment with glucose-lowering agents with proven cardiovascular benefit (IB) and receive multidimensional care, including lifestyle counseling, nutritional therapy, and self-management education and support (IC).[64]

Finally, for those with ischemic stroke or TIA who are overweight or obese, weight loss is recommended (IB), and for those who are obese, "referral to an intensive, multicomponent, behavioral-lifestyle modification program is recommended to achieve sustained weight loss" (IB).[64]

Urinary Tract Infection

UTIs are common and should be suspected with a decline in the functional status of a stroke patient, particularly in the setting of fever, leukocytosis, dysuria, or a change in continence. Initially, indwelling catheters should be removed within 24 hours of admission to rehabilitation (IB) as able. Urinalysis with reflex cultures may identify UTIs early and prevent urosepsis, but care must be taken not to overtreat as to avoid secondary consequences of antibiotic treatment, such as *Clostridioides difficile* colitis.[122]

Gastrointestinal Bleeds

A review of 6853 patients with ischemic stroke found that 100 (1.5%) developed a gastrointestinal (GI) bleed during hospitalization, with 36 (0.5%) requiring transfusion.[91] Studies are lacking in the particular population of stroke survivors in terms of gastric ulcer prophylaxis, so data from other populations must be utilized to make an informed decision on treatment. Although many physicians prescribe histamine 2 (H2) blockers with the thought that they will avoid the negative effects of proton pump inhibitors (PPIs), one review[71] of critically ill patients elicits concern related to this practice. When both enterally fed and nonenterally fed patients were included, studies demonstrated a decrease in GI bleeding rate but no demonstrable effect on hospital mortality with the use of an H2 blocker. However, in the subgroup of patients receiving enteral feeds, the patients treated with an H2 blocker demonstrated both increased mortality and a lack of protective effect against GI bleeds. It is up to the judgment of the physician to interpret this information for the individual patient. Of particular relevance to stroke treatment, the combination of a PPI and clopidogrel has not been definitively shown to increase the risk for cardiovascular events. If the combination is used, we would recommend omeprazole based on data from the COGENT trial.[114]

Pain

Central Pain

Classically seen in Déjèrine-Roussy syndrome related to thalamic stroke, neuropathic pain may occur after a stroke in a variety of locations. Appropriate care includes a thorough assessment of etiology, and other causes of pain should be excluded (IC). Medication management should be tailored to the individual, taking side effects and comorbidities into consideration (IC). Amitriptyline and lamotrigine have support as first-line agents (IIaB). Although there are less data in the literature on other tricyclic antidepressants (TCAs), consideration may be given to the use of nortriptyline or other TCAs that have fewer anticholinergic side effects than amitriptyline. Pregabalin, gabapentin, carbamazepine, and phenytoin may be considered as second-line agents (IIbB). As is frequently the case in chronic pain, an interprofessional approach should be attempted (IIaB), with utilization of behavioral interventions as well as medications and modalities through various team members. Transcutaneous electrical nerve stimulation (TENS) and deep brain stimulation have not been established as effective treatments (IIIB), but there is some evidence to support the possible use of motor cortex stimulation, if accessible (IIbB), in patients who are poorly responsive to other interventions.[122]

Shoulder Pain

Hemiplegic shoulder pain is very commonly encountered during rehabilitation and may restrict participation in and toleration of therapy. Because of the multifactorial nature of shoulder pain after stroke, a thorough examination should be undertaken—particularly including a musculoskeletal evaluation, assessment of spasticity, determination of subluxation, and peripheral neurological assessment—to determine regional sensory and motor impairment that may be outside the effect of CNS injury (IIaC). Ultrasound can be helpful for the diagnosis of soft tissue injury (IIbB).[122] Additionally, a thorough knowledge of myofascial pain referral patterns and evaluation for complex regional pain syndrome (CRPS) with compression of the metacarpophalangeal joints and evaluation for skin and autonomic changes may be helpful in identifying etiology.[112] In certain cases, electrodiagnostic evaluation can be beneficial should peripheral nerve or nerve root injury be suspected. In terms of prevention, patients and families should be trained on positioning and appropriate range of motion (IC). The strongest evidence recommends botulinum toxin for hypertonic muscles contributing to shoulder pain (IIaA), neuropathic pain medications in those with signs consistent with neuropathic pain (e.g., allodynia and dysesthesia) (IIaA), and the use of supportive devices and slings to limit forces leading to subluxation and thus also potentially decreasing pain (IIaC). Less convincing evidence also exists to support neuromuscular electrical stimulation (NMES) (IIbA), acupuncture

(IIbB), suprascapular nerve block (IIbB), and subacromial/glenohumeral joint steroid injection (IIbB).[122] Although previous work on NMES for subluxation has shown improvement in subluxation but a lack of significant improvement in pain and function,[60] new implantable devices[28] may be more useful than peripheral surface stimulation in decreasing pain, and these are beginning to be used. Should CRPS exist, a course of oral steroids, bisphosphonates, visual imagery (including mirror visual feedback), and stellate ganglion blocks may all be considered.[18] There is also an option of surgical tenotomy of shoulder internal rotators in cases of severely restricted shoulder range of motion (IIbC). Overhead pulley exercises are particularly not recommended (IIIC).[122] Finally, in the case of adhesive capsulitis, hydraulic distention with or without steroid has been used with promising but mixed results.[20] Both fluoroscopic and ultrasound-guided techniques have been used successfully.[7]

Contractures

Contractures may frequently develop after stroke and can restrict eventual function and cause potential pain and skin breakdown. Resting the hand and wearing wrist orthoses as well as regular stretching or serial casting may be helpful for reducing elbow and wrist contractures (IIbC). Surgical release of elbow flexor muscles can be considered for elbow pain and contractures (IIbB). Ankle splints may also be helpful for preventing ankle plantarflexion contractures (IIbC). Care must be taken with splinting to avoid skin breakdown. Guidelines recommend positioning the hemiplegic shoulder in the maximum external rotation tolerated for 30 minutes a day (IIaB).[122]

Skin Ulcers

Appropriate skin management requires regular assessments with use of standardized measures, such as the Braden scale (IC). Regular interventions—including nutritional assessment, skin hygiene maintenance, turning and weight-shift schedules, and the appropriate use of pressure-relieving mattresses and wheelchair cushions—are also recommended, along with ongoing family education (IC).[122]

Falls

Falls are a critical issue after stroke and require both a formal fall prevention program during inpatient rehabilitation (IA) and continued screening in the outpatient environment using an established measure, such as the Berg Balance Scale[70] or the Morse Fall Scale[81] (IIaB). Information should be provided to patients and caregivers to target environmental modification (IIaB). Participation in exercise programs with a goal to reduce falls is recommended (IB), and tai chi is a potentially effective intervention to decrease the risk of falls (IIbB).[122] One small study demonstrated an increased risk for hip fractures caused by falls in chronic elderly stroke patients deficient in vitamin D, suggesting a benefit to supplementation.[90]

Depression/Psychiatric Issues

Guidelines strongly promote standardized assessments of mood, education with a focus on adjustment, and the careful use of antidepressants in those with poststroke depression (IB). However, there is no clear recommendation of a particular antidepressant class in this population despite the fact that selective serotonin reuptake inhibitors (SSRIs) are generally well tolerated (IIIA). The use of SSRIs or dextromethorphan/quinidine is recommended for the treatment of pseudobulbar affect after stroke (IIaA). In addition to behavioral health professional consultation (IIaC), education, counseling, and support (IIbB), a consistent exercise program of at least 4 weeks may be helpful for poststroke depression (IIbB).[122]

Substance Use

2021 AHA/ASA guidelines recommend that patients with stroke or TIA be advised to stop smoking and receive counseling with or without drug therapy to assist in quitting (IA) and recommend avoidance of environmental (passive) tobacco smoke (IB). Males who consume more than two alcoholic drinks a day and females who consume more than one alcoholic drink a day should be counseled on eliminating or reducing consumption (IB). Individuals should also be counseled to avoid stimulant substances and intravenous drug abuse (IC). Those with a substance use disorder should receive specialized services (IC).[64]

Rehabilitation: Treatment of Stroke Sequelae

For the initial inpatient hospitalization, the guidelines recommend that early rehabilitation for hospitalized stroke patients should be "provided in environments with organized, interprofessional stroke care" (IA) but that "very early mobilization within 24 hours of stroke can reduce the odds of a favorable outcome at 3 months and is not recommended" (IIIA). In terms of provision of rehabilitation after the initial hospitalization, guidelines suggest that stroke patients who are "candidates for postacute rehabilitation receive organized, coordinated, and interprofessional care" (IA) and that those who "qualify for and have access to an IRF [inpatient rehabilitation facility] should receive care at an IRF rather than a SNF [skilled nursing facility]" (IB).[122]

Functional Assessment

Although a strong recommendation exists to obtain a formal assessment of functional status in both the acute hospitalization and subsequent levels of care (IB), the exact nature of the assessment is less certain. Standardized measures of motor impairment, upper extremity function, balance, and mobility are all generally thought to be possibly useful (IIbC). It is also likely that serial assessment with standardized measures could help to determine progress (IIbC). The communication evaluation should include a range of interview, observation, and both standardized and nonstandardized measures with a goal of identifying strengths and weaknesses and developing compensatory strategies (IB). All stroke patients should receive a cognitive screening (IB), and those with deficits should receive more detailed testing (IIaC). Assessment should also include sensory impairments in touch, vision, and hearing (IIaB).[122]

Cognition/Aphasia/Speech

The strongest recommendation for nondrug interventions appears to be the use of enriched environments (IA), including the use of technology (such as internet, virtual reality, and music) and other resources to improve patient engagement in the therapy process. Class IIaB recommendations exist for the provision of cognitive rehabilitation, including aspects of practice, compensation, and adaptive techniques to improve attention, memory, neglect, and executive function. There is IIbA evidence for the use of both compensatory strategies and external aids for memory. Further, IIbB evidence exists for the use of errorless learning and music therapy for memory impairments. Finally, exercise may be helpful in improving cognition and memory (IIbC).

Guidelines strongly recommend therapy for speech-language pathology (IA) and partner training (IB) for those with aphasia. Although intensive therapy is recommended, there is no agreement on the best parameters for treatment (including amount, timing, intensity, distribution, and duration) (IIaA). Care may also be supplemented with computerized treatment (IIbA). No clear guidance is given on the exact treatment approach, but group therapy may be helpful, and pharmacotherapy may be considered (IIbB).

Recommendations for motor speech disorders, including apraxia of speech and dysarthria, include techniques and strategies that target both physiological support for speech (respiration, phonation, articulation, and resonance) and global aspects of speech (loudness, rate, and prosody) (IC). Augmentative and alternative communication devices (IC) and telerehabilitation (IIaC) may be implemented as necessary. Additionally, listener education and other environmental modifications can help to promote successful communication (IIbC).

Several medications are used to treat cognitive and language impairments after stroke. Both donepezil[23] and rivastigmine[85] have shown promise in this regard, but their use is still not well established (IIbB). Additionally, antidepressant medication was shown in one study to potentially have a beneficial effect on the prevention of long-term deficits in executive function; however, an immediate effect was not noted (IIbB).[86] Atomoxetine, methylphenidate, and modafinil are used in the community, although published data are limited (IIbC).[122]

Dysphagia/Nutrition

Early dysphagia screening is important to decrease the risk of aspiration, malnutrition, and dehydration (IB). Beyond the bedside swallow assessment, an instrumental assessment is probably indicated (IIaB), but the exact type of assessment is not totally clear (IIbC). In one review,[74] incidence of dysphagia after stroke differed as follows: Cursory screening techniques identified an incidence of 37% to 45%; skilled screening identified a rate of 51% to 55%; and instrumental testing, often via videofluoroscopy, identified an incidence of 64% to 78%. The presumption is that there is a higher sensitivity with a more intensive evaluation. The incorporation of the principles of neuroplasticity (IIaB) and behavioral interventions (IIbA) is recommended for dysphagia treatment, but the use of drugs, NMES, transcranial direct current stimulation, and transcranial magnetic stimulation is not recommended in the most recent guidelines (IIIA).[122]

In the hospitalized patient with poor oral intake, feeding via nasogastric tube should be started within 7 days (IA); this should be advanced to a percutaneous G-tube in those unable to advance from tube feeding within 2 to 3 weeks after stroke (IB). Supplements may be necessary in some to prevent or treat malnutrition (IIaB). Assessment for calcium and vitamin D supplementation should be encouraged in those in long-term care facilities (IA).[122]

Sensory Impairment

Vision

In terms of eye movement impairments, guidelines strongly support the use of exercises to treat convergence insufficiency (IA). There is also some support to a lesser degree for compensatory scanning techniques both for functional activities of daily living (ADLs) (IIbB) and for scanning and reading (IIbC). For visual field deficits, yoked prisms (IIbB), compensatory training (IIbB), and computerized vision retraining (IIbC) may all be considered, although the evidence is not strong. Regarding visual/spatial and perceptual impairments, evidence for multimodal audiovisual spatial exploration training appears to be the strongest (IB). Virtual reality may also be considered (IIbB), but guidelines otherwise do not support or refute any particular intervention (IIbB). Apart from the previously mentioned use of yoked prisms, exercises for convergence insufficiency, and computer-assisted training for visual field deficits, guidelines do not support the use of other behavioral optometry methods, such as eye exercises and colored filters (IIIB).[122]

Hearing

The data are less compelling in the treatment of hearing impairments, but guidelines support referral to an audiologist for testing, use of amplification systems, use of communication strategies, and the minimization of background noise (IIaC).[122]

Hemineglect/Hemi-inattention

Multiple interventions are recommended for hemineglect and hemi-inattention, including "prism adaptation, visual scanning training, optokinetic stimulation, virtual reality, limb activation, mental imagery, and neck vibration combined with prisms" (IIaA).[122] Lower-level evidence recommends right visual field testing and repetitive transcranial magnetic stimulation, although this would likely be very difficult to implement given problems of availability (IIbB). Additionally, somatosensory retraining to improve sensory discrimination may be considered for individuals with sensory impairments (IIbB).[122]

Motor Impairment

Notably, the guidelines find that the superiority of any of the specific approaches to neurorehabilitation (e.g., neurodevelopmental therapy; Bobath, Brunnstrom, or proprioceptive neuromuscular facilitation) has not been established (IIbB).[122]

Apraxia: Evidence for the treatment of apraxia is limited, though guidelines support strategy or gesture training (IIbB) as well as task practice with or without mental rehearsal (IIbC).

Ataxia/Balance Impairment: Guidelines recommend balance screening for all individuals with stroke (IC) and strongly recommend a formal balance training program for those with poor balance and fear of falls (IA) as well as an assessment for assistive devices and orthoses in those with balance issues after a stroke (IA). Additionally, "postural training and task-oriented training may be considered for rehabilitation of ataxia" (IIbC).[122]

Hemiparesis: In terms of medications, SSRIs have the most support based on a randomized controlled trial (RCT),[25] but the guideline authors do not believe that the evidence has reached the level of a solid recommendation (IIbB), and lack of effect in subsequent studies with suggestion of increased fall risk and seizures should be taken into account.[76] Despite a small RCT showing possible short-term effect, the guideline authors do not believe that the use of levodopa is adequately established (IIbB) and, based on a negative RCT, recommend against the use of methylphenidate or dextroamphetamine for motor recovery (IIIB).[40]

Spasticity

Guidelines strongly recommend the use of botulinum toxin in both the upper and lower extremities in appropriate cases (IA). Oral medications (IIaA), such as baclofen, dantrolene, and tizanidine, and physical modalities (IIbA), such as NMES and vibration, are also recommended. In severe cases, intrathecal baclofen (IIbA) may be helpful. Current guidelines do not support the use of splints and taping for wrist and finger spasticity after stroke (IIIB).[122]

Activities of Daily Living

Guidelines recommend that patients should receive therapy that is functional and appropriately challenging and that allows for repeated practice (IA). Additionally, both ADLs (IA) and instrumental ADLs (IB) training should be tailored to the patient's needs and discharge situation. Constraint-induced movement therapy (CIMT) is recommended for those with adequate activation (IIaA). Robotic therapy and NMES are recommended for those with more moderate to severe upper limb paresis and minimal volitional movement within the first few months (IIaA). Mental practice (IIaA), strengthening exercises (IIaB), and virtual reality augmentation should also be considered as adjuncts to functional therapy. Bilateral training programs may also be useful (IIbA). Finally, acupuncture is not recommended (IIIA) because of lack of evidence that it provides benefit to improve ADLs.[122]

Mobility

The guidelines strongly recommend intensive mobility task training and the use of ankle-foot orthoses (AFOs) in those with forms of foot drop (IA). Guidelines give strong recommendations for the assessment and provision of ambulatory assist devices (IB), AFOs (IB), and wheelchairs (IC). Additionally, there is a recommendation for group therapy with circuit training and cardiovascular training (IIaA). NMES devices are recommended in certain patients with foot drop in place of an AFO (IIaA). Less strong recommendations exist for treadmill training (with or without body weight support) and robot-assisted mobility training in qualifying patients and mechanically assisted walking for those not yet able to otherwise participate in ambulation (IIbA). Guidelines do not currently support the use of acupuncture, TENS, rhythmic auditory cueing, EMG biofeedback, or water-based exercises (IIbB), but research is ongoing. There is a consideration that virtual reality may be beneficial for gait (IIbB).[122]

Bladder/Bowel

Bladder incontinence is one of the most important predictors of poorer functional outcomes, institutionalization, and mortality.[77] Although detrusor hyperreflexia is the most common subtype of incontinence after cortical and internal capsule ischemic stroke, there is a relatively higher incidence of detrusor areflexia in patients with cerebellar infarction and hemorrhagic stroke.[17] During acute hospitalization, all patients with stroke should provide a urological history and be assessed for any concern regarding urinary retention through bladder scanning or intermittent catheterizations after attempted voiding (IB). If present, a Foley catheter should be removed within 24 hours after admission (IB). Prompted voiding, through a timed voiding schedule, may be helpful in the hospital or at home, and pelvic floor muscle training may be helpful after discharge to home (IIaB).[122]

There are few high-quality studies related to bowel incontinence after stroke, but usual care involves following the frequency and consistency of bowel movements using the Bristol Stool Scale and adjusting stool softeners and laxative medications as necessary to maintain regular bowel movements.

Sexual Dysfunction

Guidelines recommend offering patients an opportunity to discuss sexual issues and concerns in the hospital and outpatient setting (IIbB).[122] Topics may include safety concerns, changes in libido, and physical and emotional consequences of stroke. Pharmacological treatment for erectile dysfunction with a phosphodiesterase-5 (PDE-5) inhibitor may be considered more than 6 months poststroke assuming there are no other contraindications or drug-drug interactions relative to the use of a PDE-5 inhibitor.[1,2]

Transition to Home and Community

With regard to transitioning from a hospital environment to the community, guidelines support an individually tailored discharge plan with the potential to utilize multiple methods of communication and support (IIaB). Caregiver involvement is thought to be important, addressing education, training, counseling, assessment, the development of support structures, and financial assistance and guidance (IIbA). Early involvement of caregivers is also supported (IIbB). In terms of resources in the community, a treating facility should provide information based on an up-to-date database on community resources, consider patient and caregiver preferences in making referrals, and follow up with patients to ensure that they are receiving appropriate and necessary services (IC).[122]

All patients should be considered for community or home-based rehabilitation (IA); when recommended, a formal plan should be developed with a case manager or other clinical staff as a point person to ensure appropriate implementation (IIbB). Additionally, caregivers should be trained in supporting and implementing the program (IIaB).[122]

Guidelines strongly recommend participation in a tailored exercise program in the home or community after the completion of formal therapy (IA) with evidence to supporting improved fitness (IA) and a reduced risk for secondary stroke (IB). Guidelines recommend promoting "engagement in leisure and recreational pursuits" through the development of self-management skills beginning as early as the inpatient environment (IIaB). For those considering a return to work, vocational rehabilitation services may be useful (IIaC). For those who return to work, an assessment of physical and cognitive abilities may be considered (IIbC).[122]

Before returning to driving, a formal on-the-road evaluation is recommended (IC). Additionally, in preparation for an on-the-road test, an assessment of cognitive, perceptual, and physical abilities is reasonable (IIaB). For those who fail an on-the-road test, a driver rehabilitation program may be considered (IIaB). Finally, the plan for a return to safe driving may include the use of a driving simulator (IIbC).[122]

Prognosis

The determination and communication of prognosis after stroke can be a very challenging task. It is important for the rehabilitation physician to correctly interpret the current data to be able to guide patients and their support systems in planning and expectations. This includes several components: (1) understanding common measures looking at outcome, (2) giving the likelihood of different overall levels of mortality and functional recovery based on initial presentation, (3) giving realistic time frames for natural recovery to occur, and (4) guiding expectations of improvement with particular impairments and activity limitations.

Outcome Measures

Common and historically well-established instruments to measure disability after stroke include the Barthel Index (BI), the Modified Rankin Scale (MRS), and the Glasgow Outcome Scale (GOS) (Tables 45.7–45.9). Familiarity with these scales can assist in interpreting studies of general stroke outcomes. When measured on admission to rehabilitation, scores on the BI[47] less than 40 tend to predict lack of independence in motor skills and difficulty with other basic skills, whereas scores over 60 tend to demonstrate a transition from dependence to assisted independence. Scores of 95 or greater have been used as a measure of complete independence in stroke outcome research.[30] The MRS is a simple 0 to 6 ordinal scale that divides subjects into general functional categories.[99] Although originally intended for brain injury, the GOS[58] is used in some studies of stroke recovery and has more recently been expanded into the Glasgow Outcome Scale–Extended.[121]

TABLE 45.7 Barthel Index

Activity	Score
Feeding: 0 = unable, 5 = needs help cutting, spreading butter, etc., or requires modified diet, 10 = independent	
Bathing: 0 = dependent, 5 = independent (or in shower)	
Grooming: 0 = needs help with personal care, 5 = independent face/hair/teeth/shaving (implements provided)	
Dressing: 0 = dependent, 5 = needs help but can do about half unaided, 10 = independent (including buttons, zips, laces, etc.)	
Bowels: 0 = incontinent (or needs to be given enemas), 5 = occasional accident, 10 = continent	
Bladder: 0 = incontinent or catheterized and unable to manage alone, 5 = occasional accident, 10 = continent	
Toilet use: 0 = dependent, 5 = needs some help but can do something alone, 10 = independent (on and off, dressing, wiping)	
Transfers (bed to chair and back): 0 = unable, no sitting balance, 5 = major help (one or two people, physical), can sit, 10 = minor help (verbal or physical), 15 = independent	
Mobility (on level surfaces): 0 = immobile or <50 yards, 5 = wheelchair independent, including corners, >50 yards, 10 = walks with help of one person (verbal or physical) >50 yards, 15 = independent (but may use any aid; e.g., stick) >50 yards	
Stairs: 0 = unable, 5 = needs help (verbal, physical, carrying aid), 10 = independent	
Total: (0–100)	

Modified from Granger CV, et al. Stroke rehabilitation: analysis of repeated Barthel index measures. Arch Phys Med Rehabil. 1979;60(1):14-17.

TABLE 45.8 Modified Rankin Scale

Score	Description
0	No symptoms
1	No significant disability despite symptoms; able to carry out all usual duties and activities
2	Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance
3	Moderate disability; requiring some help but able to walk without assistance
4	Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance
5	Severe disability; bedridden, incontinent, and requiring constant nursing care and attention
6	Dead

Modified from van Swieten JC, Koudstaal PJ, Visser MC, et al. Interobserver agreement for the assessment of handicap in stroke patients. Stroke. 1988;19(5):604-607.

Initial Presentation Prediction

In terms of initial evaluation, the NIHSS is a widely studied and utilized tool with utility in determination of prognosis, and it is recommended that rehabilitation practitioners become familiar with it to guide prognosis. Using the GOS and the BI, Adams et al. (Figs. 45.5 and 45.6) demonstrated a high probability of death or severe disability in patients with a NIHSS equal to or greater than 16 and a likely good recovery for a score equal to or less than 6.[4] Excellent recovery was noted in 46% of patients with NIHSS 7 to 10 and in 23% of patients with scores 11 to 15. Fonarow et al. found a highly predictive correlation between NIHSS and 30-day mortality in Medicare beneficiaries with ischemic stroke. Thirty-day mortality rates were 4.2% for 0 to 7, 13.9% for 8 to 13, 31.6% for 14 to 21, and 53.5% for 22 to 42.[38] Even in patients with minor stroke, 10-year mortality was seen to be 32%, a relative risk of 1.7 compared with age- and sex-matched populations.[84]

Many factors are correlated with poorer outcome after ischemic stroke, such as lesion volume and age, though neither is highly specific, and both are dependent on other factors.[119] The maximally treated intracerebral hemorrhage score[105] and other scales have been validated for determining functional prognosis after intracerebral hemorrhage and may be considered for use in medical decision making by the clinical team and family.[50] The FUNC score is a validated tool that will help predict independence at 90 days after an intracranial hemorrhage.[87]

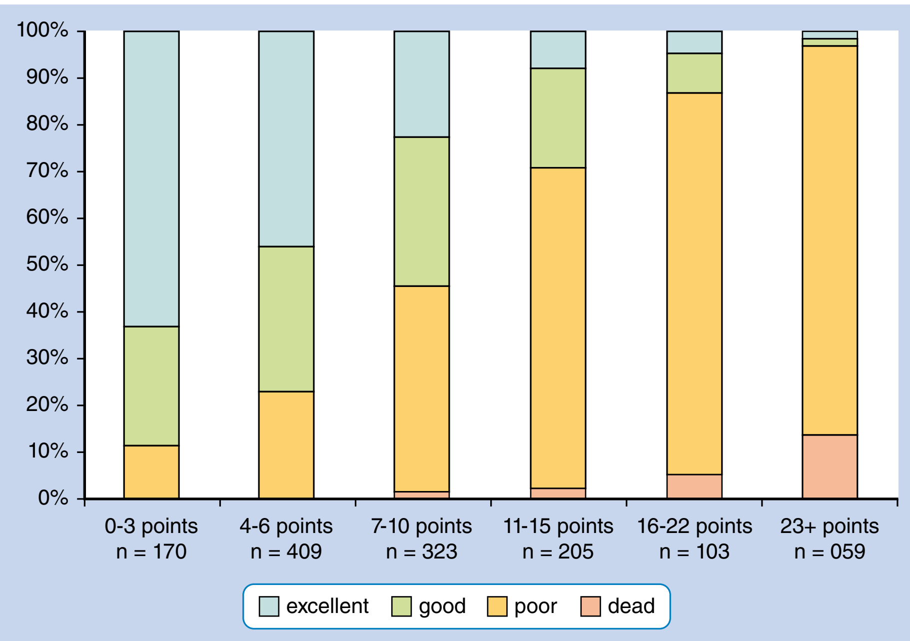
Time Frames for Recovery

For ischemic stroke, the Copenhagen Stroke Study provides guidance on time frames for duration of recovery, with mildly impaired patients requiring on average 2 months, moderately impaired patients requiring 3 months, severely impaired patients requiring 4 months, and the most severely impaired patients requiring

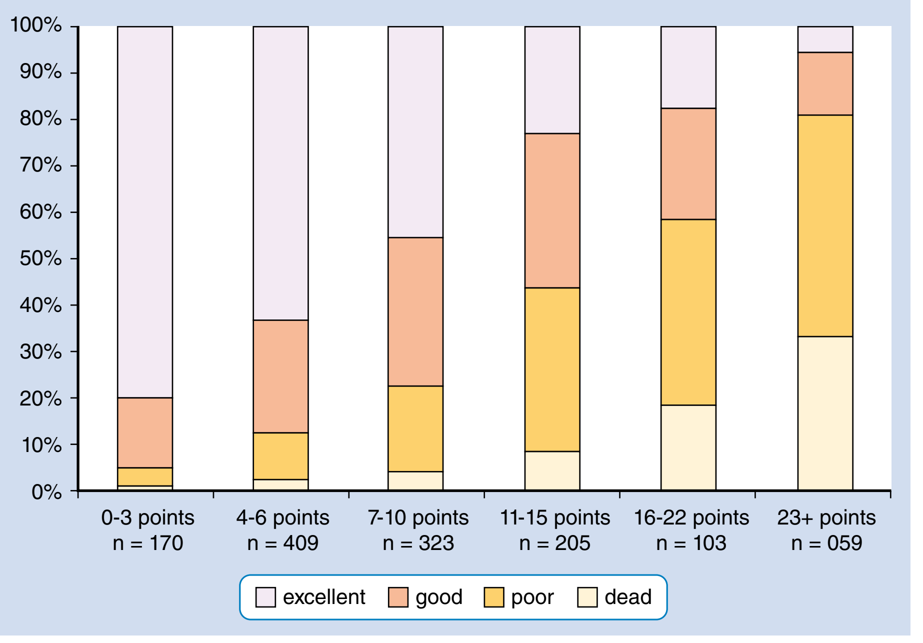
TABLE 45.9 Glasgow Outcome Scale (GOS) and Glasgow Outcome Scale–Extended (GOS-E)

GOS		GOS-E
1. Death	Patient has sustained severe injury or has died without recovery of consciousness.	1. Dead
2. Persistent vegetative state	Patient has sustained severe damage with prolonged state of unresponsiveness and a lack of higher mental functions/condition of unawareness with only reflex responses but with periods of spontaneous eye opening.	2. Vegetative state
3. Severe disability	Patient has sustained severe injury with permanent need for help with activities of daily living.	
	Patient is dependent for daily support because of mental or physical disability—usually a combination of both. Patient cannot be left alone for >8 h at home.	3. Low severe disability
	Patient is dependent for daily support because of mental or physical disability—usually a combination of both. Patient cannot be left alone for >8 h at home.	4. Upper severe disability
4. Moderate disability	Patient does not need assistance in everyday life; employment is possible but may require special equipment.	
	Patient has some disability such as aphasia, hemiparesis, or epilepsy and/or deficits of memory or personality but is capable of self-care. Patient is independent at home but dependent outside. Patient is not able to return to work even with special arrangements.	5. Low moderate disability
	Patient has some disability—such as aphasia, hemiparesis, or epilepsy—and/or deficits of memory or personality but is capable of self-care. Patient is independent at home but dependent outside. Patient is able to return to work with special arrangements.	6. Upper moderate disability
5. Low disability	Patient has sustained light damage with minor neurologic and psychological deficits.	
	Patient is able to resume normal life with the capacity to work even if preinjury status has not been achieved. Some patients have minor neurologic or psychological deficits. These deficits are disabling.	7. Low good recovery
	Patient is able to resume normal life with the capacity to work even if preinjury status has not been achieved. Some patients have minor neurologic or psychological deficits. These deficits are not disabling.	8. Upper good recovery

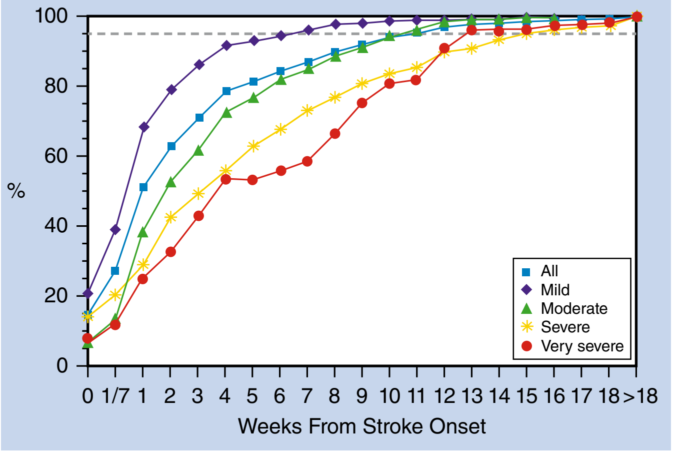
Modified from Jennett B, Bond M. Assessment of outcome after severe brain damage. Lancet. 1975;1(7905):480-484; Wilson JT, Pettigrew LE, Teasdale GM. Structured interviews for the Glasgow Outcome Scale and the extended Glasgow Outcome Scale: guidelines for their use. J Neurotrauma. 1998;5(8):573-585.



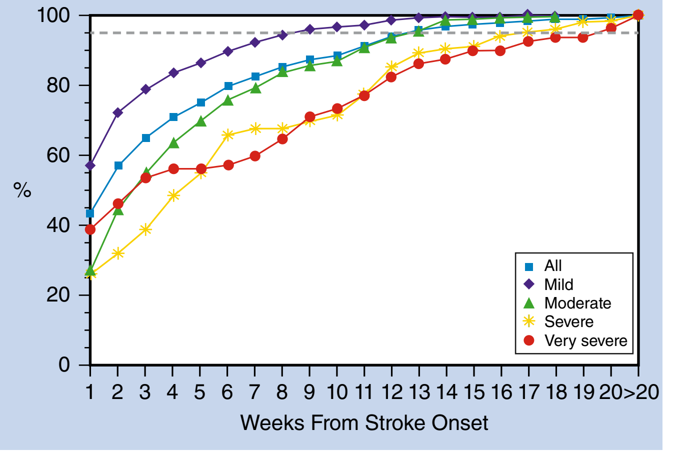
• **Fig. 45.5** Effect of baseline National Institutes of Health Stroke Scale score on outcome at 7 days. (From Adams HP Jr., Davis PH, Leira EC, et al. Baseline NIH Stroke Scale score strongly predicts outcome after stroke: a report of the Trial of Org 10172 in Acute Stroke Treatment [TOAST]. Neurology. 1999;53[1]:127.)



• **Fig. 45.6** Effect of baseline National Institutes of Health Stroke Scale score on outcome at 3 months. (From Adams HP Jr., Davis PH, Leira EC, et al. Baseline NIH Stroke Scale score strongly predicts outcome after stroke: a report of the Trial of Org 10172 in Acute Stroke Treatment [TOAST]. Neurology. 1999;53[1]:128.)



• **Fig. 45.7** Time course of recovery (peak to best neurological outcome). (From Jorgensen HS, Nakayama H, Raaschou HO, et al. Outcome and time course of recovery in stroke. Part II: time course of recovery. The Copenhagen Stroke Study. Arch Phys Med Rehabil. 1995;76[5]:408.)



• **Fig. 45.8** Time course of recovery (peak to best activities of daily living). (From Jorgensen HS, Nakayama H, Raaschou HO, et al. Outcome and time course of recovery in stroke. Part II: time course of recovery. The Copenhagen Stroke Study. Arch Phys Med Rehabil. 1995;76[5]:409.)

5 months.[60,61] Later studies have shown potential recovery even further out with severely impaired patients, with Hankey et al. showing the median time to recovery in the 18.5% of severe patients who achieved independence (an MRS score <3) in their study to be 18 months.[52] This further reinforces the need to take into account stroke severity in determining times frames of recovery in stroke (Figs. 45.7 and 45.8). Notably, discoveries in the reversal of learned nonuse and the potential of massed practice within the CIMT model have altered opinions on time frames of recovery, and patients in certain situations may be reasonably encouraged to continue to engage in attempts to regain function under the appropriate circumstances.[72] However, at the same time, it is important to note that there is a general trend of decreased quality of life and less functional independence based on a Barthel score greater than 95 during the 5 years after stroke that is independent of age, stroke severity, and other predictors of decline.[30,31]

Recovery Thresholds/Individual Impairments

Regardless of intensity of rehabilitation and therapeutic interventions, impairments and limitations are common after an ischemic cerebrovascular accident (CVA). The Framingham Study demonstrated a range of issues after first-time ischemic CVA at evaluation 6 months out from initial hospitalization: 26.2% with a BI below 60, 30.8% unable to walk unassisted, 22.2% with bladder incontinence, 35.3% with depression, and 25.9% institutionalized. Of note, the likelihood of a BI below 60 was dependent on age, including 16.7% of those aged 65 to 74, 25.5% of those aged 75 to 84, and 45% of those aged 85 to 94 years.[62]

In terms of individual impairment and disability, despite the heterogeneity of stroke presentation and outcome, there are data to guide expectations on time frames for improvement.

Aphasia recovery, similar to motor recovery, is dependent upon initial impairment, but one study suggests time to stationary language function after stroke to be 2 weeks for mild aphasia, 6 weeks for moderate aphasia, and 10 weeks for severe aphasia.[95]

Dysphagia is common in stroke patients, with up to 20% of patients in one study being shown by modified barium swallow to be proven aspirators.[95] However, despite this incidence, a relatively smaller percentage requires the eventual placement of a percutaneous gastrojejunostomy feeding tube, with only 5.7% of this same sample of stroke patients receiving a tube. Notably, risk of tube placement is much higher (19.3%) in brainstem stroke patients. In those who receive a tube, one-third are removed by discharge from rehabilitation and 75% are removed by 1 year.

For upper extremity recovery, 80% of patients achieve maximum recovery at 3 weeks and 95% achieve maximum recovery at 9 weeks.[84] Additionally, the presence of voluntary finger extension and shoulder abduction can help to guide prognosis; 98% of those with these abilities within 72 hours after stroke achieve some manual dexterity, as demonstrated by an action research arm test score of 10 or greater.[88] Among those still lacking voluntary finger extension and shoulder abduction within 72 hours, only 25% achieve manual dexterity. In those patients with a continued lack of voluntary finger extension and shoulder abduction at 9 days, eventual manual dexterity was achieved by only 14% of patients. Proprioceptive loss is also associated with decreased long-term recovery of upper extremity functioning and independence in daily living, although the difference in recovery between patients with and without proprioceptive impairment on admission to rehabilitation could not be demonstrated earlier on at 6 weeks out from stroke.[85]

In terms of predicting ambulation, sitting balance combined with ability to recruit the hemiparetic leg is a predictive marker for independent ambulation at 6 months. Of those who are able to maintain sitting balance and recruit activation in the hemiparetic leg within 72 hours from stroke, 98% were independent in ambulation at 6 months, whereas patients unable to maintain sitting balance or activate the hemiparetic leg within 72 hours had a 27% chance of independent ambulation. Those still unable to maintain sitting balance and recruit the hemiparetic leg by day 9 had only a 10% likelihood of independent ambulation at 6 months.[100]

Regarding visuospatial neglect, one study demonstrated that 31.5% of a group of 27 patients with evidence of visuospatial neglect had continued evidence of neglect at 3 months.[19] Hier et al. demonstrated a relatively more rapid recovery from left neglect, prosopagnosia, anosognosia than for hemianopia, hemiparesis, motor impersistence, and extinction.[55] The recovery from homonymous hemianopia (HH) appears to be dependent upon severity of impairment with a recovery of 17% and 72% of patients with full HH and partial HH at 1 month, respectively.[48]

Hemorrhagic Stroke

Intracerebral Hemorrhage

A recent metanalysis looking at mortality and disability after intracerebral hemorrhage showed 1- and 5-year survival rates to be 46% and 29%, respectively. Increased likelihood of death was associated with increased age, lower Glasgow Coma Scale (GCS) score, increased bleed volume, presence of intracerebral hemorrhage, and deep or infratentorial hemorrhage location. Ongoing annual risk of bleed ranged from 1.3% to 7.4% and was relatively increased after lobar rather than nonlobar bleeds. Notably, an MRS score of 0 to 2 was achieved by 53.7% to 83.7% of survivors at 6 months and 53.8% to 57.1% of survivors at 1 year. The volume of intracerebral hemorrhage is helpful in determining prognosis, with a volume of over 60 cm^3 and a GCS score below 9 showing a 91% 30-day mortality. Those with an intracerebral hemorrhage volume below 30 cm^3 and a GCS score of 9 or greater had a 30-day mortality of 19%.[82] Only 1 of the 71 patients with an intracerebral hemorrhage volume greater than 30 cm^3 could function independently at 30 days.[15] The FUNC score (an ordinal scale based on intracerebral hemorrhage volume and location, age, GCS score, and preexisting cognitive status) was shown to predict outcome.[101] In a prospective study, no patients with a score below 5 regained functional independence, whereas greater than 80% of those with a maximal score of 11 regained functional independence. Notably, 26% of the overall sample regained functional independence.

Subarachnoid Hemorrhage

Mortality is high after spontaneous SAH, with one large population study demonstrating that 10% of patients died before reaching the hospital, 25% died in the first 24 hours, and 45% died within the first 30 days. Of the 44 patients who survived 30 days, 52% were classified as "mild to no handicap," 18% as "moderate handicap" (not totally independent), 16% as "moderately severe handicap" (can be left alone for a period of time), and 14% as "totally dependent" (requiring constant supervision). Volume of bleeding is important in terms of mortality because the same study identified that only 3 of 29 patients with a SAH smaller than 15 cm^2 died within 30 days.[16,82] The International Subarachnoid Aneurysm Trial comparing coiling and clipping demonstrated a mortality of 11% at 5 years after an endovascular treatment and 14% after surgical clipping. The proportion of survivors independent at 5 years was 83% and 82% for the endovascular and clipping groups, respectively. Additionally, the standardized mortality rate of patients with ruptured aneurysms after a 1-year survival was found to be 1.57. Of the 24 rebleeds among the group of 2143 patients randomly assigned to endovascular coiling or surgical clipping, 10 were seen in the coiling group and 3 in the clipping group.[80] In one large RCT evaluating a hypothermia intervention for SAH, 873 of the 939 survivors were able to be tested via neuropsychological measures. Of those capable of being tested, cognitive impairment according to a composite score was common and is seen in up to 20% of patients, even in those making an apparently good neurological recovery, whereas impairment on at least one test was seen in 55% of subjects.[6]

Summary

In both the inpatient rehabilitation setting and the outpatient physiatric practice, any rehabilitationist caring for stroke patients must have a basic knowledge of neuroanatomy and a good understanding of the complications of stroke, including hemorrhagic transformation and acute recurrence of stroke. This person should possess the clinical tools to appropriately manage the conditions discussed in this chapter. The provider should be able to educate the patient and family on functional prognosis and the expected timeline for recovery based on published outcomes.

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